

Odoo IoT für Digital Signage

User Guide

Abilium GmbH / University of Bern (PSE FS26)

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1 Introduction

This guide explains the core features of the custom Odoo module **TRMNL**. The source code is available at:

https://github.com/Abilium-GmbH/pse_trmnl_odoo_connector

TRMNL enables Odoo to act as an API server for TRMNL e-ink displays. Users connect their devices to Odoo and show live Odoo data (calendars, tasks, sales orders, and more) on energy-efficient screens without using the TRMNL cloud.

This module was developed as part of the Software Engineering Lab at the University of Bern, in cooperation with Abilium GmbH. It is **not** affiliated with TRMNL Holdings LLC.

2 Requirements

Item	Detail
Odoo	19.0
TRMNL firmware	Custom-server firmware (e.g. v1.8.2 team-tested); the module does not enforce a minimum version.
Server	Running Odoo instance reachable from the device network.
Python packages	Pillow (required by the module manifest). The Docker <code>requirements.txt</code> may pin additional container packages.

Network: TRMNL devices must reach your Odoo server over HTTP(S). Set `web.base.url` to an address the device can use (for example `http://192.168.1.50:8069`), not `localhost` or `127.0.0.1`. Optionally set `trmnl.public_base_url` to the same LAN or public URL. During `/api/display` polls, Odoo may also use the request **Host header at runtime** to build image URLs on the device subnet, but it does **not** write that value into system parameters automatically.

Docker: The repository `README.md` covers PostgreSQL, first-time setup, and daily operation with Docker Compose.

Access: The TRMNL menu is visible to **Settings** / **Administrator** users only.

3 Device pairing

3.1 Module installation

1. In Odoo, go to **Apps**.
2. Remove the **Apps** filter from the search bar (TRMNL is classified as a module, not a standalone app).
3. Search for **TRMNL** and install the module.

3.2 Device configuration (hardware)

1. Turn on the device (switch on the back).
2. Press and hold the button on the back for **5-10 seconds**. When you release it, the screen should refresh and show instructions to connect to the TRMNL Wi-Fi network.

3. On your phone or laptop, connect to the **TRMNL** Wi-Fi network. A device configuration page should open in the browser.

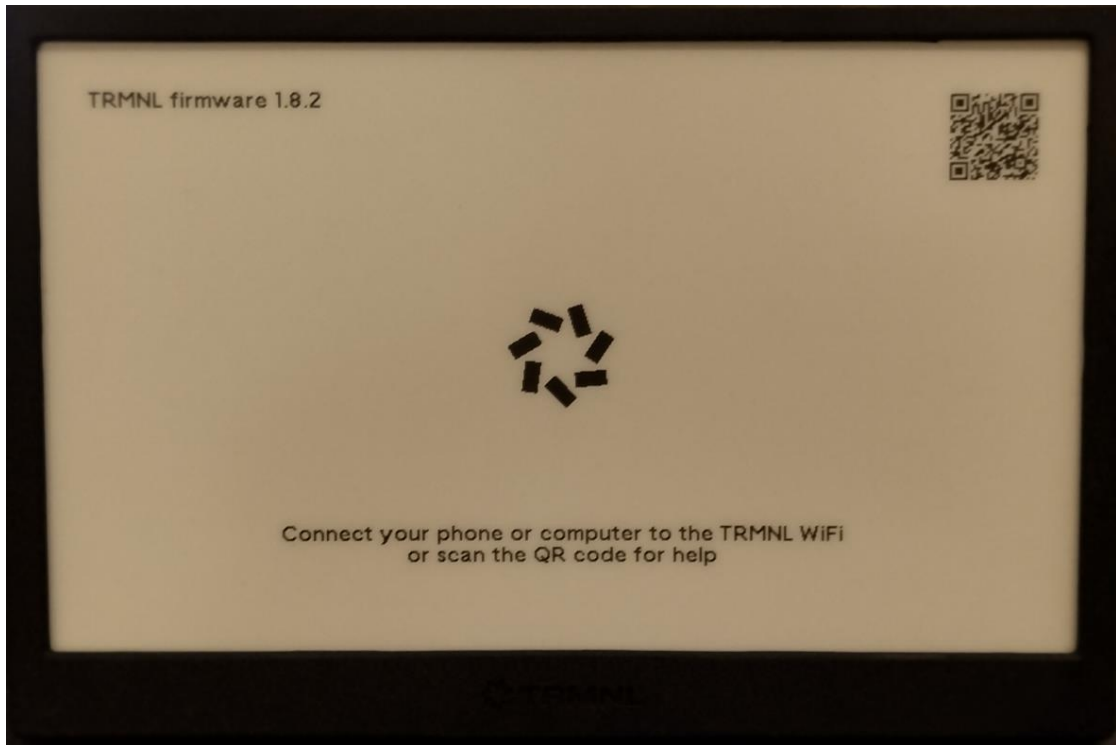


Figure 1: TRMNL setup mode: connect to the device Wi-Fi network.

3.3 Connect to your Odoo server

To use this module, clear any stored TRMNL cloud credentials first:

1. On the configuration page, scroll down and open **Advanced**.
2. Click **Soft Reset** and confirm **Yes**. This removes stored credentials and is required before pointing the device at a custom server.
3. The device may disconnect during the reset. Reconnect to the TRMNL network and return to **Advanced Configuration**.
4. Click **Custom Server** and confirm **Yes**.
5. Enter your server address:
 - **Odoo.sh:** your instance public HTTPS URL, e.g. `https://<your-instance>.odoo.com`
 - **Local / Docker:** LAN IP and port of the Odoo host, e.g. `http://192.168.1.50:8069`
6. Click **Back to Wi-Fi**.
7. In **Wi-Fi Configuration**, confirm **API Server** is visible, select your Wi-Fi network, enter the password, and click **Connect**.

Tips:

- The connection can be unstable; you may need to reconnect to the TRMNL network manually.

Wi-Fi Configuration

RVN-79102	2.4G	
JOOWIN-2G&5G	2.4G	
DIRECT-B9-HP Smart Tank 7300	2.4G	
RajNet	2.4G	
Kesar Rest	2.4G	
Comedor.v	2.4G	
Sunrise_1102471	2.4G	
dormitorio 2.ynm,	2.4G	
Du kommst hier nicht rein	2.4G	

SSID

Enter your Wi-Fi name

Password

Enter your Wi-Fi password

Connect

Refresh

Advanced

MAC: 80:F1:B2:80:52:48

Figure 2: Device configuration portal: open Advanced.



Advanced Configuration

Soft Reset

Custom Server

Run Sensor Test

Back to Wi-Fi

Figure 3: Soft Reset confirmation dialog.



Advanced Configuration

API Server

Enter your custom server without a trailing / (e.g. https://trmnl.app)

Soft Reset

Custom Server

Run Sensor Test

Back to Wi-Fi

Figure 4: Custom Server: enter your Odoo server URL.

Figure 5: Wi-Fi Configuration with API Server field visible.

- A temporary message to visit **trmnl.com/start** is hard-coded in the firmware and can be ignored. Wait for an automatic refresh or press the back button to refresh manually.
- After the first successful screen refresh, if you see the **Abilium** logo, setup succeeded and the device is ready to poll Odoo.

4 Device management

4.1 Access

Open **TRMNL** in the main menu.

Menu	Purpose
TRMNL → Devices	Registered displays, telemetry, refresh rate, and logs.
TRMNL → Profiles	Configure which Odoo data is rendered and displayed on each device.
TRMNL → Display Policy	Configure the global behavior for unknown or mismatched devices.

4.2 How devices appear in Odoo

When a TRMNL is configured with your Odoo server and connects over Wi-Fi, it uses these HTTP endpoints:

You **cannot** create devices manually from the list (**New** is disabled). Records appear when hardware contacts the server.

Depending on **Display Policy** (see below), a device that polls **/api/display** **before** a successful **/api/setup**, or with a wrong token, may appear as **Unknown Device** or **Token Mismatch** until you accept it (or until **Automatically accept** promotes it on that poll).



Figure 6: Temporary `trmnl.com/start` message (safe to ignore).



Figure 7: Successful setup: default Abilium welcome screen on the device.

TRMNL Devices Profiles Display Policy								
Devices Search... 1-1 / 1								
☐	Device Name	MAC Address	Status	Image Filename	Last Seen At	Refresh Rate (min)	Battery Percentage	Signal Strength
☐	Office Display	94:A9:90:8B:2A:A0	Accepted	profile_28_20260531T2233...	Jun 1, 12:43 AM	1	12	Good

Figure 8: TRMNL main menu in Odoo.

Endpoint	Method	Purpose
/api/setup	GET	First registration. Odoo creates a device record, marks it as Accepted , and returns a new API token.
/api/display	GET	Periodic polls for the image URL, filename, and refresh rate.
/api/log	POST	Telemetry and firmware logs. Stored on the device Logs tab only for Accepted devices with a valid token; otherwise HTTP 401 is returned and logs are discarded.
/api/profile/image/...	GET	Downloads the profile PNG image. Access is secured using an access token in the URL.

With **Automatically accept**, a device can become **Accepted** from /api/display alone, without ever calling /api/setup (the presented token is stored as the device token).

A device in **Token Mismatch** state keeps receiving the error image until you **Accept Device** (under the error policy) or switch to **Automatically accept**. Sending the previously accepted token again does not restore normal display while the record stays in that state.

If a device has no **active** profile, or Odoo cannot build a profile image URL, the display falls back to the default welcome or unauthorized image (for example the Abilium screen after setup).

4.3 Device list view

TRMNL → Devices shows:

Column	Description
(drag handle)	Manual sort order (sequence). Drag rows to reorder.
Device Name	Friendly label (editable on the form).
MAC Address	Hardware ID (read-only). Format: AA:BB:CC:DD:EE:FF .
Status	Badge: Accepted (green), Token Mismatch (yellow), and Unknown Device (red).
Image Filename	Current filename sent to the device (read-only). TRMNL firmware re-downloads the PNG only when this filename changes, not on every poll.
Last Seen At	Last contact with any TRMNL API endpoint.
Refresh Rate (min)	How often the server tells the device to poll (editable, 1–30 minutes; default 1).
Battery %	Last reported battery level.
Signal Strength	Wi-Fi quality from RSSI (Excellent , Good , Fair , or Poor).

Search filters: Accepted, Needs Attention (token mismatch or unknown device), group by Status.

List actions (select rows → **Action** menu):

Action	Effect
Accept Selected	Manually accepts devices that have presented a token. Skips devices already marked as Accepted or devices without a stored presented token. Intended for the Respond with an error policy; the header Accept Device button is shown only when that policy is active.
Remove Selected	Deletes device records from Odoo immediately.
Reset Selected	Schedules a factory reset on the next device poll and then removes the device record from Odoo.

4.4 Device form view

Open a device from the list for the full form.

Header buttons:

Button	When to use
Accept Device	Visible when Display Policy is Respond with an error and the device is not yet Accepted . Adopts the token the device last presented.
Reset Device	Schedules a factory reset on the next poll, then removes the Odoo record.
Remove Device	Removes the Odoo record only, or Reset & Remove to reset the physical device first (recommended).

The screenshot shows the TRMNL interface for a device named 'Display'. The top navigation bar includes 'TRMNL', 'Devices', 'Profiles', and 'Display Policy'. Below the navigation bar, there are buttons for 'Reset Device' and 'Remove Device'. The device status is 'Accepted', with options for 'Token Mismatch' and 'Unknown Device'. The form fields are as follows:

CONFIGURATION		TELEMETRY	
Device Name	Display	Battery Percentage	0
MAC Address	94:A9:90:8B:2A:A0	Signal Strength	Good
Image URL	http://192.168.1.127:8069/api/profile/image/17v=8cf3ffb16146&access_token=qoXybkHUQvPjn7UZ2iyXoPq68EJL3_APCF8q1vyXmc	Last Reported Refresh Rate (min)	1
Refresh Rate	1 minutes	Added At	Jun 1, 1:06 PM
Image Filename	profile_1_20260601T112654_8cf3ffb16146	Last Seen At	Jun 1, 1:34 PM
		Last API Call	Display

Below the form fields, there is a 'Logs' tab. The logs table has columns: Device Log ID, Name, Created At, Wi-Fi Status, Refresh Rate, and Battery Voltage.

Figure 9: TRMNL Devices form view with status and telemetry columns.

Form sections:

- **Configuration** (top): image URL, refresh rate (minutes), filename.
- **Telemetry**: battery, signal, last reported refresh rate, registration and last-seen timestamps.
- **Logs** (notebook tab): entries from `POST /api/log` for **Accepted** devices only (select and delete individual lines if needed).

5 Display Policy

TRMNL → **Display Policy** configures how Odoo responds to `/api/display` requests from:

- devices not yet registered via `/api/setup`, or
- devices whose **Access-Token** does not match the stored token.

It does **not** change normal operation after a successful `/api/setup` (those devices are **Accepted**

with a server-issued token).

The setting is stored in the system parameter `trmnl.display_unknown_device_policy`.

Policy	Behavior
Respond with an error (default, recommended)	The device receives an Unauthorized placeholder image. Odoo may create or update an Unknown Device or Token Mismatch record. Accept the device manually when ready.
Automatically accept the device	Odoo adopts the token from that poll, marks the device as Accepted , and serves the normal display payload.
Send a reset command	Odoo returns <code>{"status":500}</code> to the device (firmware factory-reset path) and then deletes the device record. Reconfigure Wi-Fi and the Custom Server afterwards.

This is different from the admin **Reset Device** button on the device form, which sets `reset_firmware:true` on the next poll and then removes the record.

Use **Accept Device** (or **Accept Selected**) when the policy is **Respond with an error** and a token has been stored from a prior poll.

6 Profile management and rendering

6.1 Overview

A **Profile** defines what Odoo data is drawn into a PNG for a TRMNL device:

1. Choose an **Odoo model** (e.g. Calendar Events, Tasks, Sales Orders).
2. Pick a **View Type** (layout).
3. Configure filters, fields, and layout-specific options.
4. Odoo renders an e-ink PNG (default canvas **800x480**; device-reported size is used when available). **List** layouts are binarized to 1-bit; kanban, calendar, and graph layouts use grayscale.
5. On each device poll, Odoo serves that image via a secured URL.

Each profile is linked to **one device**. A device can have **multiple profiles**; only the **lowest sequence** among **active** profiles is used for display. Reorder profiles in the list with the drag handle.

6.2 Creating a profile

TRMNL → Profiles → New

6.2.1 Identity

Field	Description
Name	Label (e.g. “Office calendar”).
Device	Target TRMNL.
Active	Inactive profiles are ignored for display.

6.2.2 Data source

Field	Description
Odoo Model	Business model to display (wizards and technical models are hidden).
View Type	Layout; options depend on the model (see below).
Users (optional)	Affects Assigned to Me (<code>user_id</code> filter) and <code>uid / user</code> in Domain Filter evaluation. Empty = current Odoo user when you click Render Preview ; on device polls the environment user is often Public , so set Users for production profiles. Does not change calendar or footer timezone (see below).

Click **Render Preview** (header) to generate the first PNG without waiting for a device poll.

The screenshot shows a web interface for managing profiles. At the top, there's a navigation bar with 'TRMNL', 'Devices', 'Profiles', and 'Display Policy'. Below this, a 'New' button is visible. The main form is titled 'IDENTITY' and contains fields for 'Name', 'Device', and 'Active' (a toggle switch). Below the identity section is the 'DATA SOURCE' section, which includes a dropdown for 'Odoo Model'. At the bottom, there's a 'DEVICE DELIVERY STATUS' section with fields for 'Device Last Polled', 'Next Expected Poll', and 'Device Refresh Rate'.

Figure 10: Profile form: identity and data source.

6.3 View types (layouts)

View Type	Use for	Key options
List	List of records	Fields to Display: first field = title, others = metadata; Max Records ; Sort Order
Kanban	Cards grouped by stage	Kanban Stage Field ; Fields to Display: first field = card title
Calendar	Month or week calendar	Calendar View , Week Mode , Reference / Reference Date
Graph	Bar or line chart	Graph Type , group-by / measure fields, Chart Title , sort and limits

View types not supported by the chosen model are not offered in the dropdown (based on installed Odoo views).

Important: The **Calendar** renderer only runs for the `calendar.event` model. If you pick another model and **Calendar** as view type, Odoo falls back to an **empty list** image. The

TRMNL module depends only on **base** and **web**; install the Odoo **Calendar** app separately and use **Calendar Event** for calendar layouts.

6.3.1 Calendar settings (Calendar layout only)

Field	Options
Calendar View	Month, Week
Week Mode	Work Week (Mon-Fri), Full Week (Mon-Sun); week view only
Reference	Current (today), Custom Date
Reference Date	Shown when Reference = Custom Date

Calendar notes:

- **Quick Filter** is hidden in the UI for calendar layouts; the month/week window comes from **Calendar Settings** and **Reference** only. **Domain Filter** still applies. **Assigned to Me** still applies to calendar loads when configured (via `user_id` on events).
- **Timezone:** calendar week view, timed event labels, and the PNG footer use the **profile creator's** Odoo timezone (`create_uid.tz`), falling back to UTC. Device polls run as the **Public** user, so the **Users** field does not control timezone. Create or recreate the profile while logged in as a user whose timezone matches what you want on the device.
- **Month, Reference = Current** uses the **server date** (`date.today()`) for the grid and the “today” cell highlight, not the profile creator timezone.
- **Week view** includes only non-all-day timed events, drawn between **07:00 and 19:00** (local time after UTC conversion using the profile creator timezone). All-day events are **not** shown. Events outside the hour grid are clipped on screen.
- **Month view** indexes events by **start** day; multi-day events appear on the start day only. Only a limited number of events fit per cell (**+N more** for overflow).
- **Max Records** applies to calendar event search as well (calendar loader falls back to **200** only when the limit is unset/zero).

6.3.2 Graph settings (Graph layout only)

Bar chart

- **Group By** field (required)
- **Measure** (blank = count records)
- **Sort** order
- **Max Bars** (default 10, max 20)

Line chart

- **Date Field** (required)
- **Measure** (blank = count)
- **Line Date Group By** (Day / Week / Month)
- **Max Points** (default 12, max 52)

Chart Title: optional; defaults to the group-by or date field label.

6.4 Filters and sorting

Field	Description
Quick Filter	No Filter, Assigned to Me, Today, This Week, This Month, Overdue (not used for the calendar date window)
Domain Filter	Extra Odoo domain (AND with quick filter). Supports <code>uid</code> , <code>context_today()</code> , <code>True</code> , <code>False</code>
Sort Order	e.g. <code>date_deadline desc</code> , <code>name asc</code> (not for graph layout)
Max Records	Cap for list/kanban ORM search (default 20). Calendar uses the same field with fallback 200 . Not shown for graph layout.

6.5 Preview and delivery

After **Render Preview** or an automatic render on poll:

- The **Preview** section shows the PNG in the form (same bytes the device downloads).
- **Device Delivery Status** shows **Device Last Polled**, **Next Expected Poll**, **Device Refresh Rate** (stored value in **seconds**, read-only), and **Image URL** when a reachable URL could be built from stored parameters (without poll Host context). The field may stay empty on the form even though the device receives a URL on the next poll.
- The footer on the final PNG shows `device_name · Last update: DD.MM.YYYY, HH:MM` in the **profile creator's** Odoo timezone (or UTC). Footer text appears only after the device has polled at least once (`last_poll_at` is set).

Render Preview

June 2026

Mon	Tue	Wed	Thu	Fri	Sat	Sun
1	2 06:30 Meeting 1	3 05:45 Meeting 2	4 08:00 Meeting 3	5	6	7
8	9	10	11	12	13	14
15	16	17	18 08:00 Meeting 4	19	20	21
22	23	24	25	26	27	28
29	30					

Display · Last poll: 2026-06-01 13:34

DEVICE DELIVERY STATUS

Device Last Polled ?	Jun 1, 1:34 PM
Next Expected Poll	Jun 1, 1:35 PM
Device Refresh Rate ?	60
Image URL	http://192.168.1.127:8069/api/profile/image/1

Figure 11: Profile preview and Device Delivery Status on the profile form.

Setting	Where	Meaning
Refresh Rate	TRMNL → Devices	How often the device polls <code>/api/display</code> (1–30 minutes; default 1).
Render Interval	Optional column on the profile list	How often Odoo regenerates the PNG (default 10 minutes) if data has not changed.

6.5.1 Refresh rate vs render interval

The device may poll every minute but still show the same PNG until a re-render is triggered (see below).

6.6 Image URLs and `web.base.url`

TRMNL devices download images over HTTP. URLs must be reachable from the device Wi-Fi network.

Situation	What happens
<code>web.base.url</code> is a LAN or public address the device can reach	Image URLs work on preview and poll; no warning.
<code>web.base.url</code> is localhost / loopback	A yellow URL Warning appears on the profile form until you set a reachable <code>web.base.url</code> and/or <code>trmnl.public_base_url</code> . A poll may still serve a working URL to the device via the Host header, but the warning can remain.
<code>trmnl.public_base_url</code> is set	Used as the base URL when building image links without poll context. The warning is suppressed if the URL is reachable.
Poll Host header on <code>/api/display</code>	Used at runtime to select a reachable base URL on the device LAN. The value is not saved to Odoo system parameters.

Recommended before pairing (Docker / local):

1. **Settings → Technical → System Parameters**
2. Set `web.base.url` to your host LAN URL, e.g. `http://192.168.1.50:8069`
3. Optionally set `trmnl.public_base_url` to the same value

On **Odoo.sh**, use your instance HTTPS URL; the device must reach it over the internet.

Profile image URLs include an `access_token` query parameter so only registered devices can download PNGs. Administrators can open the same URL when logged into Odoo without the token.

6.7 When Odoo re-renders the image

A new PNG is built when any of the following is true (on the next poll, or when you click **Render Preview**):

1. No preview exists yet
2. Source records changed (create / edit / delete on the profile model)
3. **Render interval** elapsed (default 10 minutes)
4. The **trmnl** module was upgraded (renderer version changed)

The device **poll interval** is separate: if the device polls every 15 minutes, it still receives the **latest stored** PNG even if that PNG was rendered 5 minutes ago.

6.8 End-to-end workflow (example)

1. Pair the device (see **Device pairing**) → **Accepted** after `/api/setup`, default Abilium screen.
2. **TRMNL** → **Profiles** → **New** → model **Calendar Event**, **View Type Calendar**, **Device** = your TRMNL.
3. **Render Preview** → confirm the preview in the form.
4. Wait for the next device poll (or set **Refresh Rate** to **1** minute on the device for testing).
5. The display updates to your calendar PNG.

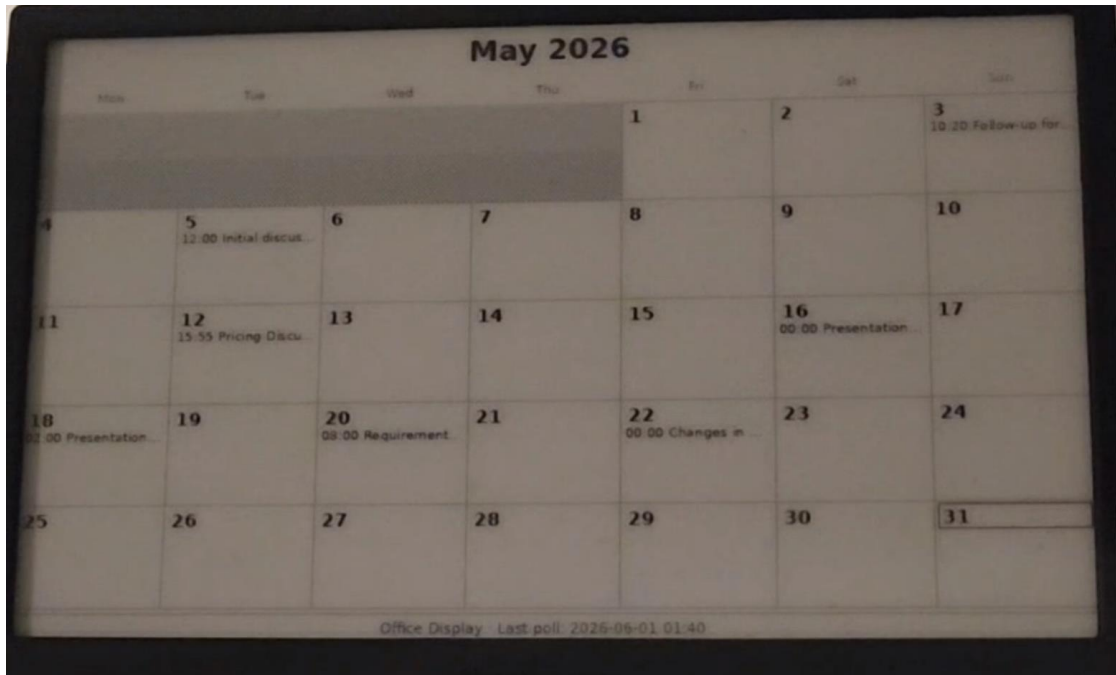


Figure 12: Calendar profile rendered on the TRMNL device.

7 Troubleshooting

7.1 Device does not appear in Odoo

Check	Action
Custom Server URL	Must match Odoo reachable address (LAN IP or Odoo.sh HTTPS), not localhost.
Soft Reset	Clear old credentials before Custom Server .
Wi-Fi	Device must join your network; reconnect to the TRMNL AP if needed.
Setup completed	Device should call <code>/api/setup</code> once; check Last Seen At on TRMNL → Devices .
Firewall	Allow inbound HTTP(S) to Odoo from the device subnet.

7.2 Device shows Unauthorized or Abilium error image

Check	Action
Display Policy	Default is Respond with an error until you Accept Device .
Status	Unknown Device or Token Mismatch → Accept Device if a token was stored.
Wrong server	Factory-reset device, re-enter Custom Server URL.
Token mismatch	Accept Device (stores the token the device last sent) or Reset Device and re-pair. Status stays Token Mismatch until accepted.

7.3 Device logs missing

Check	Action
Device status	/api/log accepts data only for Accepted devices with a matching token.
HTTP 401	Unknown device, token mismatch, missing token, or wrong token: logs are discarded. Accept the device first.

7.4 Profile preview works but device image does not update

Check	Action
URL Warning on profile	Set <code>web.base.url</code> and/or <code>trmnl.public_base_url</code> manually in System Parameters. Polls do not persist the Host header or clear the warning.
Image URL empty on form	Normal when stored base URLs are loopback; the device may still get a URL on poll. If the device keeps the old image, fix base URLs and wait for a new filename after re-render.
Wrong profile	Use lowest sequence among active profiles for that device.
Refresh Rate	Lower to 1 minute for testing; press device back button to refresh.

7.5 Empty or wrong data on the display

Check	Action
Quick Filter / Domain Filter	Too restrictive: widen or clear domain.
Users on profile	Affects Assigned to Me and <code>uid</code> in domains.
Max Records	List/kanban default 20; increase if rows are missing.
View Type	Calendar layout is intended for Calendar Event .
Render Preview	Validates Odoo rendering before blaming the device.

7.6 Calendar-specific

Symptom	Likely cause
Event missing in week view	All-day event (excluded), outside 07:00-19:00 , or filtered out.
Event on wrong day	Set the Timezone on the user who created the profile (Settings → Preferences), or use Custom Date . Users on the profile does not fix calendar TZ.
Calendar layout empty list	Model is not Calendar Event .
“+N more” on month cells	More events than fit in the cell.

7.7 Display Policy: factory reset loop

If the device keeps resetting, **Display Policy** may be **Send a reset command**. Change to **Respond with an error** or **Automatically accept**, then re-pair.

7.8 Logs and debugging

- **TRMNL** → **Devices** → device form → **Logs** tab.
- Odoo server log: search for TRMNL during setup and display polls.
- Optional: `trmnl.api_debug = True` in system parameters for verbose API logging.

7.9 Module uninstall fails

If uninstall errors mention missing columns on `trmnl_profile` (for example `active`), use a build that includes the data-watcher uninstall fix, then retry uninstall.

8 Further reading

Document	Content
README.md	Docker quick start, project overview
design_documentation.md	Architecture, HTTP API, security, data model
development.md	Make targets, tests, contributor workflow
repository_structure.md	Folder layout

9 Development team

Role	Name
Key Account Manager	Timur Umut Turgul
Chief Deliverable Officer	Sascha Friedli
Quality Evangelist	Leila Ayinkamiye
Master Tracker	Claudio Berger